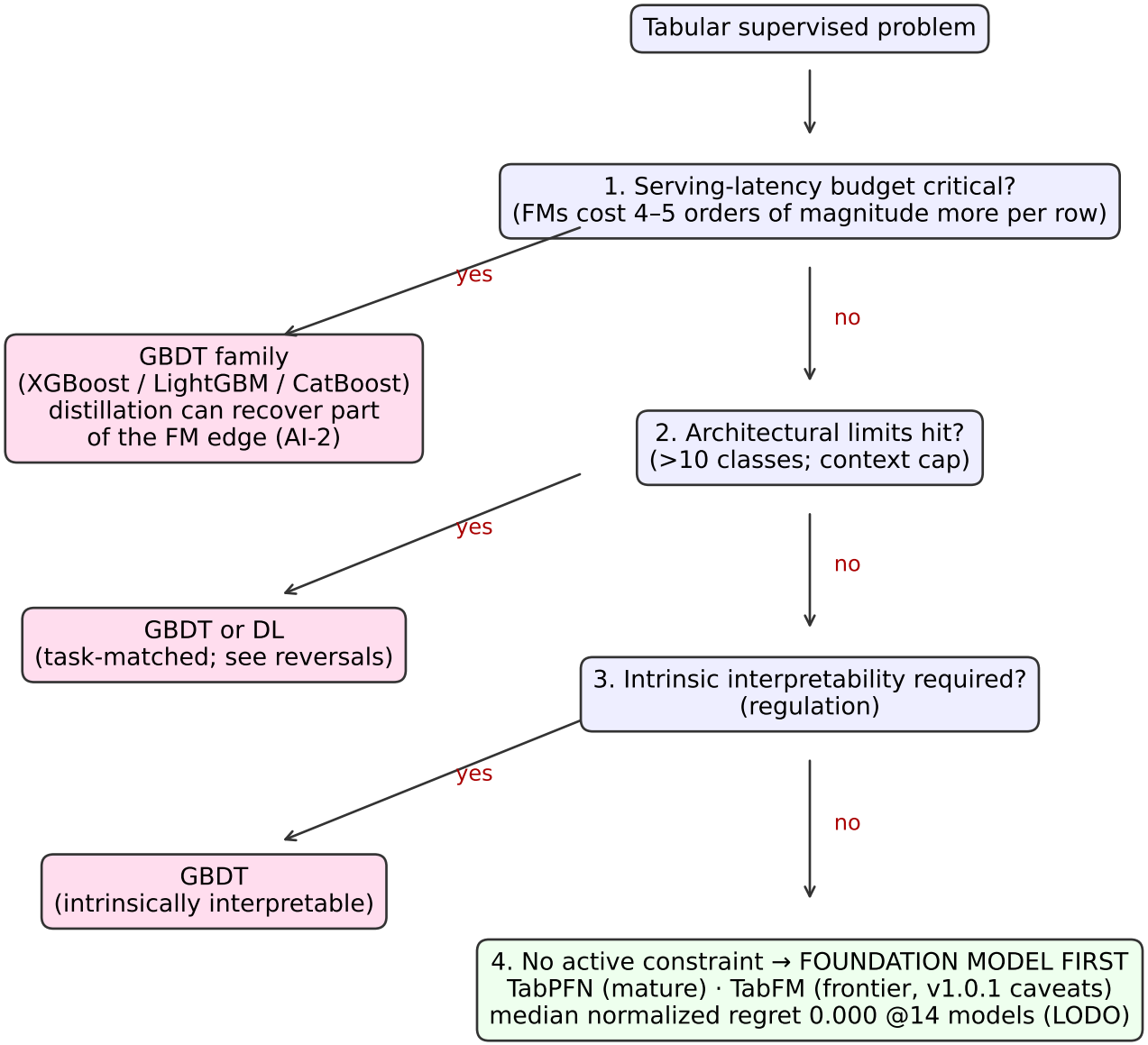
Tabular supervised problem
1. Serving-latency budget critical?
(FMs cost 4–5 orders of magnitude more per row)
yes
GBDT family
(XGBoost / LightGBM / CatBoost)
distillation can recover part
of the FM edge (AI-2)
no
2. Architectural limits hit?
(>10 classes; context cap)
yes
GBDT or DL
(task-matched; see reversals)
no
3. Intrinsic interpretability required?
(regulation)
yes
GBDT
(intrinsically interpretable)
no
4. No active constraint → FOUNDATION MODEL FIRST
TabPFN (mature) · TabFM (frontier, v1.0.1 caveats)
median normalized regret 0.000 @14 models (LODO)